

A Revised Ground-Motion Prediction Model for Shallow Crustal Earthquakes in Italy

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Abstract This work aims to revise the [Bindi *et al.* \(2011\)](#) ground-motion model for shallow crustal earthquakes in Italy (hereinafter, ITA10), calibrated in the magnitude range 4.0–6.9 using strong-motion data recorded up to the 2009 L'Aquila sequence. The improvement of ITA10 is needed because of the large number of strong-motion records made available in Italy after the occurrence of the most recent seismic sequences (2012 Emilia, Northern Italy; 2016–2017 Central Italy). The new data collection allows us to extend the magnitude range beyond 6.9 and to include vibration periods up to 10 s. Instead of the geometric mean of the horizontal components of ground motion, the median of orientation independent amplitudes (RotD50) is selected as a measure of the ground-motion parameters, and the rupture distance is introduced as an alternative source-to-site metric to the Joyner–Boore distance (R_{JB}). The site effects are accounted for by a linear dependence on the time-averaged shear-wave velocity in the upper 30 m, V_{S30} . A breakdown of the ground-motion variability is performed into between-event and site-to-site components to make the model suitable for the evaluation of nonergodic probabilistic seismic hazard. We also build a heteroscedastic model for aleatory variability as a function of moment magnitude and V_{S30} . The evaluation of the epistemic uncertainty in the median prediction is also provided to be introduced in the logic trees for the probabilistic seismic hazard assessment. We obtain changes in median predictions with respect to ITA10 at distances lower than 10 km and for strong events ($M_w > 6.5$); moreover, the total standard deviations are significantly lower at intermediate and long periods, with an average reduction of about 20%.

Electronic Supplement: Tables listing events used for the calibration of ITA18 model, coefficients of the ITA18 model for Joyner–Boore distance and rupture distance, and epistemic uncertainty of the median prediction and figures showing magnitude scaling against residuals corrected for the attenuation, distance scaling against the residuals corrected for the source and site terms, site terms as a function of V_{S30} against the residuals corrected for source and distance terms, residuals of the ITA18 model, and event- and site-corrected residuals

Introduction

Ground-motion prediction equations (GMPEs) are used in seismic hazard analysis to evaluate the expected median level of ground shaking and its associated uncertainty at any given site for a given earthquake scenario. The GMPEs are calibrated by regression of empirical or simulated ground-motion amplitudes against a set of predictor variables such as earthquake magnitude, source-to-site distance, and local soil conditions. The predicted median values are generally controlled by the choice of the functional form, adopted to simplify the complex physical process governing the ground motion. The Next Generation Attenuation-West2 (NGA-

West2) project ([Bozorgnia *et al.*, 2014](#)) developed a suite of GMPEs to be implemented in probabilistic seismic hazard assessment (PSHA) for shallow crustal earthquakes in active tectonic regions, with more than 20 explanatory variables. When applied in Italy, several input parameters such as the basin depth are challenging to define.

Regarding the assessment of GMPE uncertainties, these can be broken down into two components: (1) the inherent randomness, referred to as aleatory variability, and (2) the epistemic uncertainty, relating to a lack of knowledge ([Toro *et al.*, 1997](#)). In the current practice of PSHA, logic

tree is the standard tool used for capturing and representing the epistemic uncertainty for ground-motion prediction (Kulkarni *et al.*, 1984) and should capture both the best estimates of what is known and the potential range of alternatives in light of what is currently not known (Douglas, 2018). In the more recent backbone approach (Atkinson and Adams, 2013), the uncertainty in the median prediction of each GMPE is statistically calculated based on the model fit and the data distribution (Al Atik and Youngs, 2014) and scaled by factors consistent with the quantified epistemic uncertainty.

About the aleatory variability (sigma), several research projects (e.g., probabilistic seismic hazard analysis for the Swiss nuclear power plant sites [PEGASOS] project funded by Swiss Nuclear) focused on its proper assessment because it has a significant impact on PSHA of critical infrastructure (design for long return periods). A way to treat the sigma consists in the identification of the components of ground-motion variability that are repeatable rather than purely random so that these may be removed from the aleatory component and transferred to the quantification of the epistemic component by relaxing the ergodic assumption (Anderson and Brune, 1999; Atkinson, 2006; Al Atik *et al.*, 2010; Rodriguez-Marek *et al.*, 2011; Stewart *et al.*, 2017).

Moreover, in many GMPEs, the aleatory variability is assumed to be homoscedastic, that is, independent of the variables of the predictive model (Strasser *et al.*, 2009). Several authors (e.g., Sadigh *et al.*, 1997; Ambraseys *et al.*, 2005) have related the sigma to one or more explanatory variables and therefore suggested heteroscedastic models that commonly reduce the scatter at increasing magnitude (e.g., Bommer *et al.*, 2007). The NGA-West2 project introduced the heteroscedastic sigma in several prediction models, which are basically dependent on magnitude or, in some cases, on other explanatory variables (Abrahamson *et al.*, 2014; Boore *et al.*, 2014; Campbell and Bozorgnia, 2014; Chiou and Youngs, 2014).

The research described in this article builds on the model developed by Bindi *et al.* (2011; hereinafter, ITA10) for shallow crustal earthquakes in Italy. ITA10 consists of a set of predictive equations derived from 769 records (by 150 recording stations) of 99 earthquakes in the magnitude range 4.1–6.9, which occurred in Italy in the time span 1976–2009.

Although the overall good performance of ITA10 for shallow crustal earthquakes in active tectonic regions (Delavaud *et al.*, 2012; Lanzano *et al.*, 2016), the authors are aware of several limitations: (1) the magnitude range ($4.1 \leq M_w \leq 6.9$) limits the maximum usable magnitude to 6.9 for PSHA studies; (2) the longest period used for spectral ordinates predictions is 2 s because the ITA10 dataset contained a significant amount of analog records (about half of dataset); (3) the distance metric is only that introduced by Joyner and Boore (1981; R_{JB}), that is, the shortest distance from a site to the surface projection of the rupture; and (4) the site response is evaluated only for the Eurocode 8 (EC8) site categories (Eurocode 8, 2003).

A significant improvement of the ground-motion model proposed in this article involves the enlargement of the dataset to magnitudes larger than 6.9 to ensure robust magnitude scaling. To this goal, we add some global events in the magnitude range 6.1–8.0. In addition, the large number of digital data, available after major seismic sequences in Italy (Emilia 2012; central Italy 2016–2017), allows us to extend the period range up to 10 s. The fault-plane solutions (from Regional Centroid Moment Tensor and Centroid Moment Tensor) and 3D fault geometries, from specific literature studies, allows us to classify all the events according to their focal mechanism and calculate the distances from the rupture plane (R_{rup}) other than the R_{JB} and calibrate two alternative models. Finally, the availability of site-related studies (Felicetta *et al.*, 2017; Zimmario *et al.*, 2018) allows us to introduce the average shear-wave velocity in the uppermost 30 m (V_{S30}) as a variable to explain the site response.

Our strategy for calibrating new GMPEs for Italy (hereinafter, ITA18) is to improve the predictions both in terms of median, epistemic uncertainty, and aleatory variability while maintaining the simplicity of the functional form adopted by Bindi *et al.* (2011).

Dataset

The working dataset is extracted from the flatfile (Lanzano *et al.*, 2018), derived from the Engineering Strong-Motion (ESM) database (Luzi, Puglia, Russo, *et al.*, 2016; Luzi, Puglia, Russo, and ORFEUS WG5, 2016), including the records of some regional small-magnitude events ($M_w < 4.0$), available in the Italian Accelerometric Archive (ITACA) database (Pacor *et al.*, 2011; Luzi, Pacor, and Puglia, 2017). To increase the maximum usable magnitude with respect to ITA10, the records of 12 worldwide (global) events (3 Turkey, 2 Japan, 2 New Zealand, 2 California, 1 Iceland, 1 Iran, and 1 Greece) in the magnitude range 6.1–8.0 have been added. The selection of global earthquakes allows us to extend the magnitude range of strike-slip (SS) and thrust (TF) mechanisms that are less frequent in Italy (the maximum moment magnitudes for thrust and SS events are 6.4 for 1976 Friuli and 6.0 for 1978 Patti earthquakes, respectively). The highest moment magnitude for shallow normal fault (NF) events is 6.9, as in ITA10, and corresponds to the 1980 Irpinia earthquake because we could not retrieve any global event with larger magnitude that was adequately sampled and well documented. To avoid oversampling of events that could be not representative of the regional attenuation or stress drop, we include only a small percentage of worldwide earthquakes (8% of the total number of events), corresponding to the 14% of the dataset.

The complete list of events is provided in © Table S1 (available in the electronic supplement to this article) and includes the event ID (ESM/ITACA ID or IDs from international catalogs), location, focal depth, moment magnitude, style of faulting (SoF), and number of records.

The strong-motion data selection is performed according to the following constraints:

- *Crustal conditions*: Earthquakes of active shallow (event depth < 30 km) crustal regions; different regimes are excluded (volcanic events or subduction events in the southern Tyrrhenian sea).
- *Magnitude*: Moment magnitude range is from 3.5 to 8.0, in which the maximum magnitude for Italian events is 6.9 (1980 Irpinia earthquake).
- *Spatial distribution*: Not all the events with magnitudes less than 5.0 are included to avoid oversampling of small magnitudes; several aftershocks of major seismic sequence in Italy are disregarded to have, as much as possible, a homogeneous spatial distribution of events.
- *Event sampling*: Events having less than 10 records are excluded.
- *Distance*: Source-to-site distances lower than 200 km are included.
- *Components of ground motion*: Only records having three components (two horizontal and one vertical) are considered; the vertical component is also included for future studies on the same dataset.
- *Soil–structure interaction*: Only surface instruments with low or no interactions with nearby structures are included. However, most of the Italian stations managed by the Department of Civil Protection are installed in small masonry buildings that usually host electric devices; thus, we exclude several recording sites for which the hosting structure can significantly affect the records (Ditommaso *et al.*, 2010).

The final selection includes 5607 records, relative to 146 earthquakes and 1657 stations. The moment magnitude of the events is attributed according to European-Mediterranean Earthquake Catalogue (EMEC; Grünthal and Wahlström, 2012), which is included in the ESM flatfile. If the events are not in the EMEC catalog, we assign moment magnitudes following this hierarchy: (1) literature studies, (2) Regional Centroid Moment Tensor (Pondrelli and Salimbeni, 2015) and Centroid Moment Tensor (Ekström *et al.*, 2012), and (3) Time-Domain Moment Tensor (Scognamiglio *et al.*, 2009).

For events with magnitudes larger than 5.5, the rupture geometries are defined, and the finite-fault distances are calculated. For smaller magnitude events ($M_w < 5.5$), we consider point-like sources because the differences between the epicentral and R_{JB} as well as between the hypocentral and rupture distances can be neglected. About 500 stations (30%) are characterized by measured V_{S30} , and in case of missing data, they have been inferred from slope, according to the procedure by Wald and Allen (2007).

The events are mapped in Figure 1, in which the regional earthquakes are represented as a function of magnitude (Fig. 1a) and SoF (Fig. 1b). Global events are represented as a function of SoF in Figure 1c. Earthquakes with magnitudes larger than 5.5 are mainly located in central Italy and the predominant SoF is normal because the seismicity of the

Italian peninsula is dominated by the tectonic extension along the Apennine chain.

Figure 2a shows the magnitude–distance distribution of the records included in the ITA18 dataset compared with those used for the calibration of ITA10. The calibration dataset is about seven times larger, resulting in a significant improvement of the records sampling in near-source conditions and in the magnitude range 5.0–6.5. In particular, the amount of near-source records in ITA18 consists of ~300 records with distances lower than 10 km. Figure 2b shows the magnitude–distance distribution (R_{JB}) of the regional versus worldwide records. Figure 2c,d represents the number of records in ITA18 and ITA10 as a function of distance and magnitude. The new dataset is balanced across all distances and in the magnitude range 3.5–6.5 (> 600 samples in each bin). The amount of records used for ITA18 calibration is at least two times larger than ITA10 in each magnitude and distance bin.

Data Processing and Ground-Motion Intensity Measures

The records are uniformly and manually processed following the procedure of Paolucci *et al.* (2011). The amount of digital data is strongly increased, thanks to the installation of new recording stations and the replacement of analog instruments. A huge number of records are also available after the installation of temporary networks during the recent seismic sequences. As a result, the quality of records is considerably improved, and the number of records per event is significantly larger, leading to more than 100 records within 200 km for the most recent seismic sequences in Italy.

Because the processing is manual, high-pass filter corner frequencies may differ. As a result, the number of usable records varies with periods (Boore and Bommer, 2005) and reduces from about 5600 at 0.1 s to about 4100 at 10 s.

The model is calibrated for peak ground acceleration (PGA) and peak ground velocity (PGV) and for 36 ordinates of acceleration response spectra (SA) at 5% damping in the 0.01- to 10-s period (T) range. The prediction is valid for RotD50, which is the median of the distribution of the intensity measures (IMs), obtained from the combination of the two horizontal components across all nonredundant azimuths (Boore, 2010). We do not include equations for peak ground displacement (PGD) because the adopted processing procedure cannot capture the PGD when the records exhibit permanent displacement in the near-source region. In the latter case, different processing schemes should be adopted (e.g., Kamai, Abrahamson, and Graves, 2014; D'Amico *et al.*, 2018).

Calibration of the Predictive Model

Predictor Variables

The predictor variables are the moment magnitude M_w , the source-to-site distance (either the closest distance to the

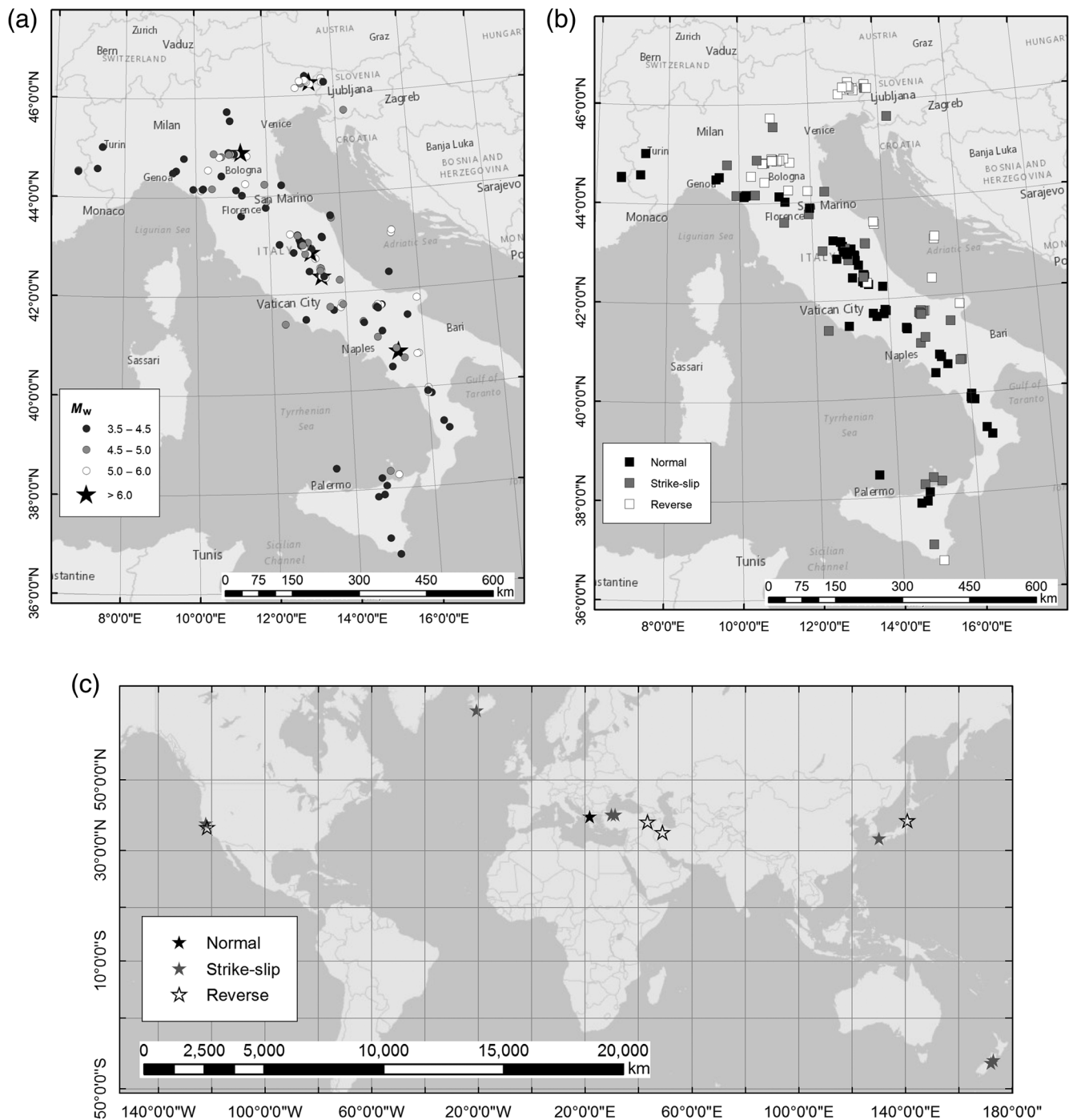


Figure 1. Map of the events used to calibrate ITA18 (see © Table S1, available in the electronic supplement to this article). Italian earthquakes as a function of (a) moment magnitude, (b) style of faulting (SoF), and (c) worldwide earthquake ($M_w \geq 6.0$) as function of SoF.

rupture plane R_{rup} , or the Joyner–Boore distance R_{JB}), and the V_{S30} as site parameter. We also account for the SoF, representing the classification of events as SS, TF, or NF, based on the rake angle (Boore and Atkinson, 2008).

Figure 3 shows the attenuation with distance of SA ordinates at two periods ($T = 0.1$ and 10 s) for seven magnitude classes. The magnitude dependence on attenuation is evident at short periods because a slope change at magnitude around

5.5 can be recognized. On the contrary, the slope variation at long periods (e.g., $T = 10$ s) is much slighter. Similar conclusions can be drawn for the magnitude scaling, shown in Figure 4. At short periods (e.g., $T = 0.1$ s), the magnitude scaling is affected by a large variability, especially at smaller magnitudes, and can be assumed bilinear with a slope break in the magnitude range between 5.0 and 6.0. At long periods, the SA ordinates linearly scale with magnitude almost in the

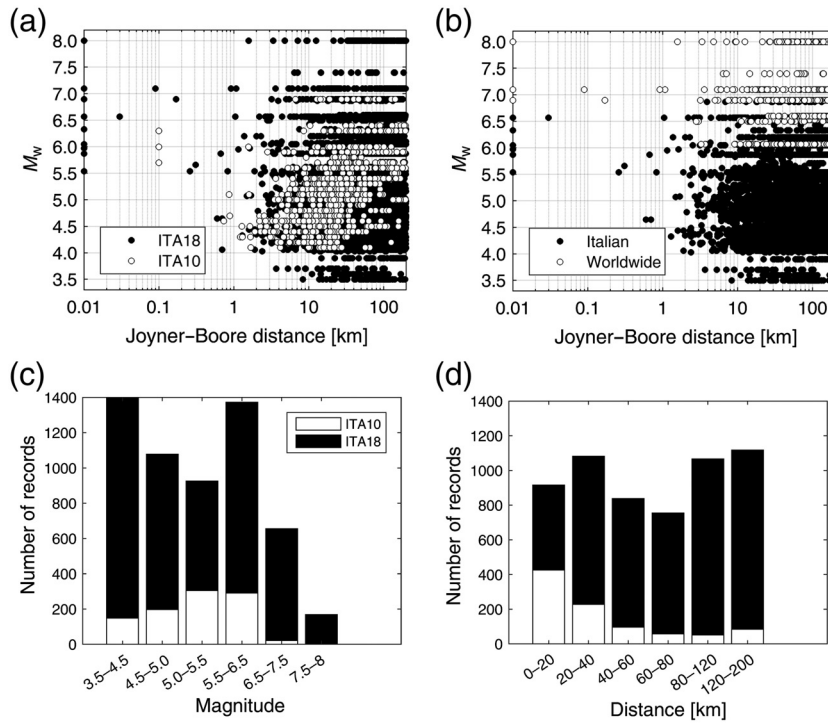


Figure 2. Magnitude distance distribution of records. (a) ITA18 versus ITA10. (b) Italian versus global records in ITA18 dataset. Histograms of records as a function of (c) moment magnitude and (d) Joyner–Boore distance.

entire range (slope break for $M_w \geq 7.0$) with lower scatter. According to Figure 4, we set the magnitude slope in the GMPE functional form to be stepwise linear.

Functional Form

The general expression adopted for the new set of GMPEs (ITA18) is modified after [Bindi et al. \(2011\)](#)

$$\log_{10} Y = a + F_M(M_w, \text{SoF}) + F_D(M_w, R) + F_S(V_{S30}) + \varepsilon, \quad (1)$$

in which Y is the observed IM, a is the offset, SoF is the style of faulting, $F_M(M_w, \text{SoF})$ is the source function, $F_D(M_w, R)$ is the distance function, $F_S(V_{S30})$ is the site term, and ε is the error associated with the median prediction.

The source is modeled as two terms:

$$F_M(M_w) = \begin{cases} b_1(M_w - M_h) & \text{where } M_w \leq M_h \\ b_2(M_w - M_h) & \text{where } M_w > M_h \end{cases}, \quad (2)$$

$$F_M(\text{SoF}) = f_j \text{SoF}_j. \quad (3)$$

Different from [Bindi et al. \(2011\)](#), we adopt a linear function for $M_w \leq M_h$. Whereas coefficients b_1 and b_2 control the magnitude scaling, the SoF coefficients f_j s provide the correction for focal mechanism. We set the magnitude slope to be stepwise linear, and we assume M_h as the period-dependent hinge magnitude. In equation (3), SoF_js are dummy

variables, introduced to specify SS ($j = 1$), reverse TF ($j = 2$), and normal NF ($j = 3$) fault types, and f_j are the SoF coefficients (f_1 for SS, f_2 for TF, and f_3 for NF). The regression is performed constraining to zero the coefficient for normal faulting ($f_3 = 0$).

The path function has the form

$$F_D(M_w, R) = [c_1(M_w - M_{\text{ref}}) + c_2] \log_{10} R + c_3 R, \quad (4)$$

in which the first term is the magnitude-dependent geometrical spreading and the second is the anelastic attenuation, M_{ref} is the reference magnitude, and c_1 – c_3 are the path coefficients. $R = \sqrt{R_i^2 + h^2}$, in which i can be either rup to indicate rupture distance, and JB indicates Joyner–Boore distance; h is the pseudodepth (km) and is estimated by the regression.

Finally, the site term is a function of V_{S30}

$$F_S(V_{S30}) = k \log_{10} \left(\frac{V_0}{800} \right), \quad (5)$$

in which $V_0 = V_{S30}$ when $V_{S30} \leq 1500$ and $V_0 = 1500$ m/s otherwise. Different from ITA10, we select V_{S30} as the variable representing the site amplification because it is more flexible than other proxies such as the EC8 site categories, being unrelated to a specific seismic code. We assume that the site term scales linearly with V_{S30} with respect to a reference velocity of 800 up to 1500 m/s because 800 m/s can be considered representative for rock sites in Italy ([Norme Tecniche per le Costruzioni \[NTC\], 2018](#)). Because the record sampling of very hard-rock sites is poor, we set an upper bound on the V_{S30} scaling, corresponding to 1500 m/s, above which the amplification is independent on V_{S30} ([Kamai, Abrahamson, and Silva, 2014](#)).

Step I: Evaluation of the Fixed Parameters

We perform a nonlinear regression to evaluate the hinge magnitude M_h in equation (2), the reference magnitude M_{ref} , and the pseudodepth h in equation (4). Because the hinge magnitude M_h is found to be fairly stable in the regression, we assume values in the 5.5 (short periods)–6.3 (long periods) range, similar to [Boore et al. \(2014\)](#), regardless of the adopted distance metric. M_{ref} is found to be dependent on distance metrics and spectral periods. At longer periods, M_{ref} tends to assume lower values, down to 4.0. The pseudodepth h is dependent on the distance metric and assumes values in the 3.3- to 7.2-km range. In general, h -values are lower than in [Bindi et al. \(2011\)](#), and the general effect on median predictions is an increase of the ground motion at shorter distances. M_h , M_{ref} , and h can be found in © Table S2.

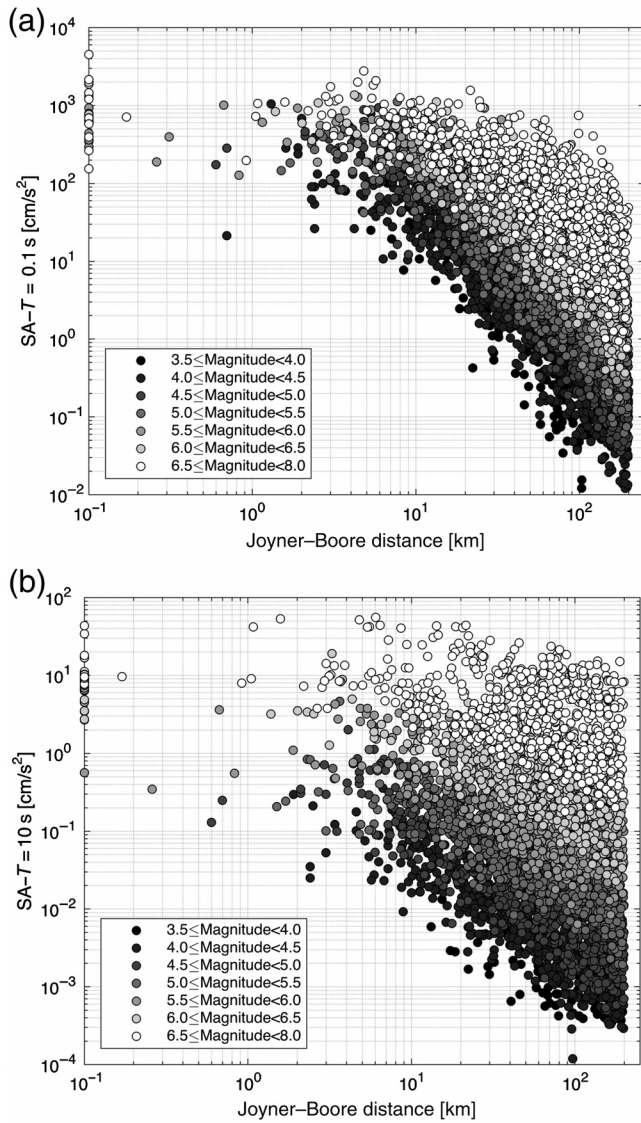


Figure 3. Attenuation with distance of the observed intensity measures (IMs) for different magnitude groups. (a) $T = 0.1$ s and (b) $T = 10$ s.

Step II: Linear Regression with Mixed Effects

In the second step, we perform a linear ordinary least-squares mixed-effect regression (Bates *et al.*, 2015) to evaluate source, path, and site term coefficients, setting the period-dependent parameters of step I. The random effects are applied to stations and events to estimate the partially nonergodic sigma according to Al Atik *et al.* (2010). The total standard deviation σ can be calculated as:

$$\sigma = \sqrt{\tau^2 + \phi_{S2S}^2 + \phi_0^2}, \quad (6)$$

in which τ and ϕ_{S2S} represent between-event and site-to-site variability, respectively, and ϕ_0 is the standard deviation of the event- and site-corrected residuals. GMPE coefficients and homoscedastic values for ϕ_0 , τ , and ϕ_{S2S} are available in © Table S2 for R_{JB} and R_{rup} . The coefficient for anelastic

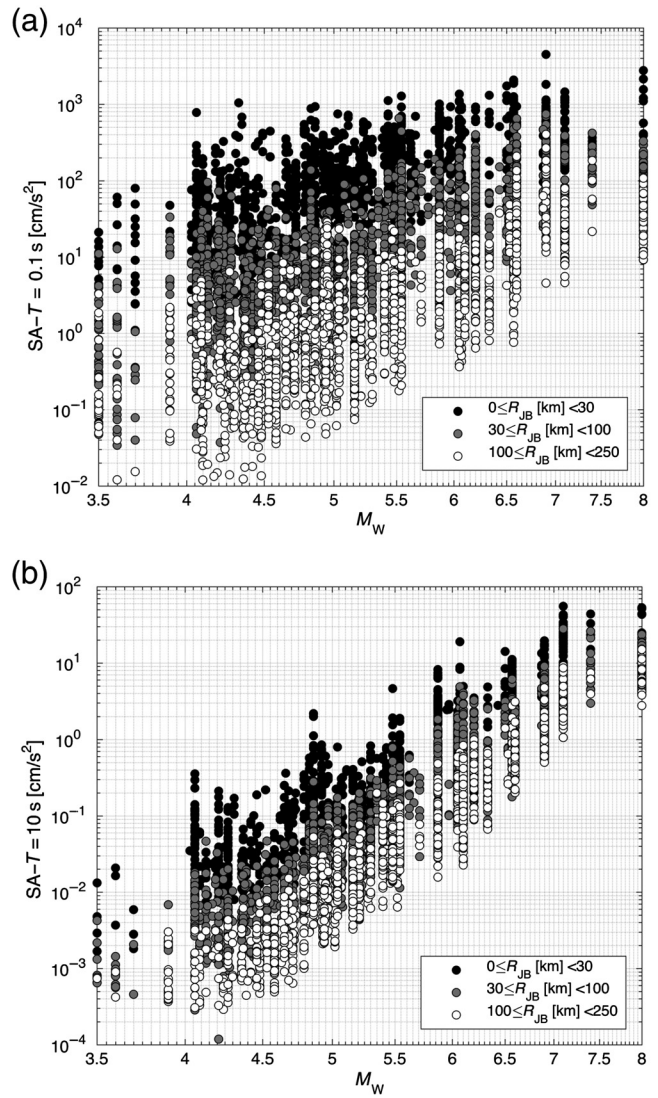


Figure 4. Magnitude scaling of observed IMs for different distance groups. (a) $T = 0.1$ s and (b) $T = 10$ s.

attenuation c_3 tends to assume positive values, even if very small, at intermediate and long periods, leading to an enhancement of the spectral amplitudes with respect to logarithmic decay. This is corrected by setting to zero the value of c_3 in the regression, when c_3 is positive.

Tables 1 and 2 report the correlation matrices of the GMPE coefficients at $T = 0.1$ and 10 s, respectively. At short periods, the largest negative trade-offs are observed between c_2 and c_3 and c_2 and a ; the highest positive trade-off is between a and c_3 . This evidence is similar to that of Bindi *et al.* (2017) and reflects the trade-off between source and attenuation terms. At long periods, the largest positive trade-off occurs between a and b_1 , but significant negative values are found between b_1 and b_2 and c_1 and c_2 .

The statistical significance of the predictors is tested through the p -value (Wasserstein and Lazar, 2016). A low p value (< 0.05) indicates that the null hypothesis can be rejected. In other words, a predictor that has a low p value

Table 1
Correlation Matrix of the ITA18 Coefficients for Acceleration Response Spectra (SA) at $T = 0.1$ s

Coefficients	a	b_1	b_2	c_1	c_2	c_3	k	f_1	f_2
a	1								
b_1	0.6119	1							
b_2	-0.3977	-0.4256	1						
c_1	-0.0216	-0.4476	-0.5961	1					
c_2	-0.8877	-0.3135	0.2065	-0.0467	1				
c_3	0.8827	0.3137	-0.1952	0.036	-0.9995	1			
k	0.0991	-0.1385	-0.0284	0.1673	-0.0548	0.043	1		
f_1	-0.2886	0.0181	-0.3961	0.2484	0.1567	-0.1559	-0.1191	1	
f_2	-0.4674	-0.23	-0.1435	0.2061	0.2375	-0.2373	-0.0813	0.6288	1

Symmetric

Table 2
Correlation Matrix of the ITA18 Coefficients for SA at $T = 10$ s

Coefficients	a	b_1	b_2	c_1	c_2	k	f_1	f_2
a	1							
b_1	0.9241	1						
b_2	-0.6212	-0.6989	1					
c_1	-0.1532	-0.2557	-0.4649	1				
c_2	-0.4068	-0.198	0.5199	-0.6947	1			
k	0.0708	-0.0944	-0.0516	0.2335	-0.2591	1		
f_1	-0.1042	0.1194	-0.5882	0.4819	-0.2486	-0.1092	1	
f_2	-0.464	-0.2392	-0.2097	0.3981	-0.062	-0.1183	0.6585	1

Symmetric

is likely to be a meaningful addition to the model because changes in the predictor's value are related to changes in the response variable. In Table 3, the p value and the square of the correlation between the response values and the predicted response values (R^2) are shown for some IMs (PGA, PGV, and SA at $T = 10$ s). As expected, the p value is higher than 0.05 for b_2 at short periods because of the large variability in magnitude scaling and uncertainty in M_h estimation. The coefficients of SoF correction f_1 and f_2 , exhibit higher p values, showing that these terms are not a meaningful addition to the model. This result is in line with those observed by Bommer *et al.* (2003) since the introduction of this fourth explanatory variable in functional form (out of magnitude, distance, and V_{S30}) has a small (less than other variables) or negligible effect on standard deviation of the GMPEs. As a trial test, we calibrate a model without a classification scheme for SoF, and as a result, the other GMPEs coefficients ($a, b_1, b_2, \dots, k$) and the standard deviations are almost identical to ITA18. Following Bommer *et al.* (2003), we decide to keep the SoF classification in the functional form of the GMPE despite its limited impact on model prediction.

Finally, R^2 close to 1 indicates that a greater proportion of variance is accounted for by the model. In this regression, R^2 is larger than 0.96 at all IMs.

Interpretation of Regression Results

In this section, we discuss the regression results for source, distance, site terms, and residuals for the ITA18, R_{JB} , and

Table 3

Statistical Test for ITA18: p Value for Each Coefficient; Square of the Correlation between the Response Values and the Predicted Response Values (R^2)

Statistical Test	Coefficients	PGA	PGV	SA - $T = 10$ s
p Value	a	0	0	$\ll 0.05$
	b_1	$\ll 0.05$	$\ll 0.05$	$\ll 0.05$
	b_2	0.593	< 0.05	< 0.05
	c_1	$\ll 0.05$	$\ll 0.05$	$\ll 0.05$
	c_2	0	0	0
	c_3	$\ll 0.05$	< 0.05	–
	k	$\ll 0.05$	$\ll 0.05$	$\ll 0.05$
	f_1	< 0.05	0.200	0.688
	f_2	0.760	0.687	0.909
R^2		0.964	0.968	0.979

PGA, peak ground acceleration; PGV, peak ground velocity.

ordinates of SA at 0.1 and 10 s. SA at $T = 0.1$ s is selected because it is the spectral ordinate having the largest variability and can be considered as a proxy of PGA. SA at $T = 10$ s is relevant to show that the model can be successfully extended to very long periods. However, the results for two widely used parameters for engineering applications (PGA and PGV) are shown in © Figures S1–S5.

Analysis of the Source, Path, and Site Terms

The magnitude function F_M includes two linear terms intersecting at the hinge magnitude M_h . In particular, a linear

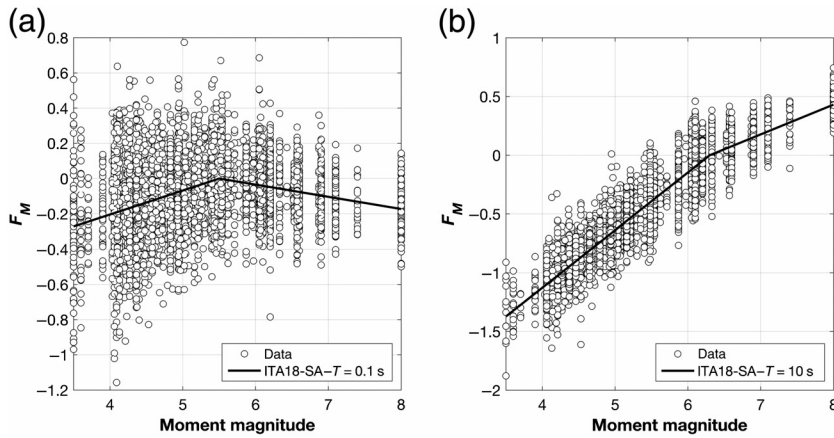


Figure 5. Magnitude scaling against residuals corrected for the attenuation and site term. (a) $T = 0.1$ s and (b) $T = 10$ s.

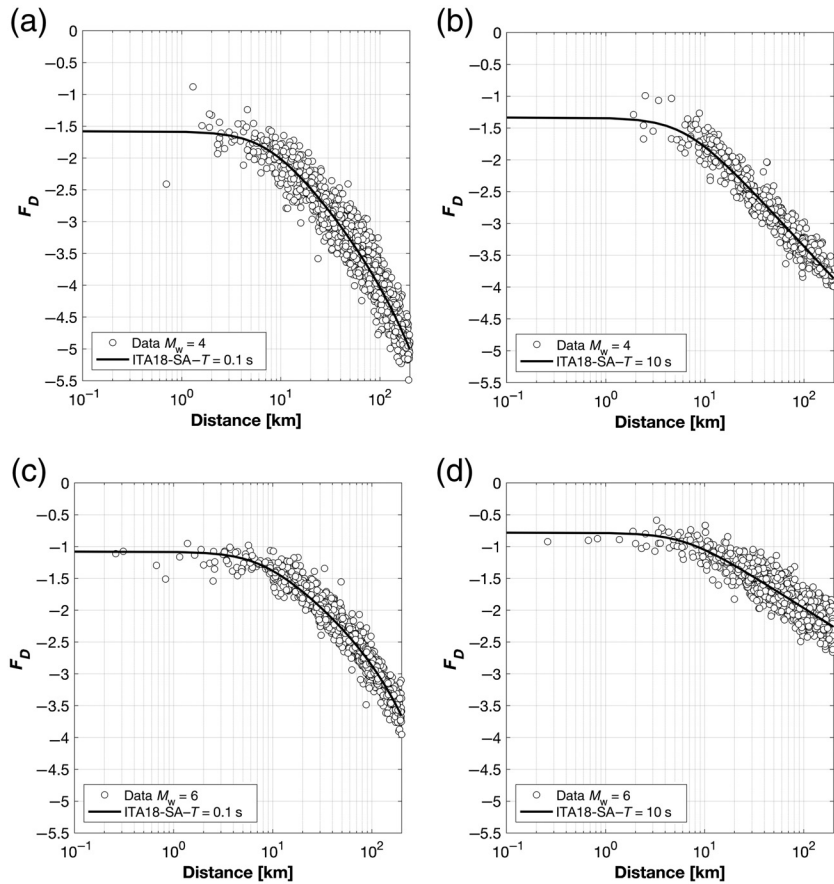


Figure 6. Distance scaling against the residuals corrected for the source and site term (white circles). (a) $T = 0.1$ s and M_w 4.0, (b) $T = 10$ s and M_w 4.0, (c) $T = 0.1$ s and M_w 6.0, and (d) $T = 10$ s and M_w 6.0.

model is introduced for $M < M_h$ differently from [Bindi *et al.* \(2011\)](#) because statistical tests to assess the performance of the quadratic model gave p -values larger than 0.05, indicating low statistical significance of this term.

We do not constrain the magnitude slope to be positive (b_1 and b_2 coefficients in $\textcircled{E}$ Table S2), although sometimes

it results in negative slopes for short-period IMs after the hinge magnitude break (Fig. 5a). Although the negative slopes may suggest oversaturation, they do not cause a decrease of the IMs at increasing magnitude because even at zero distance, the magnitude-dependent geometrical spreading compensates the oversaturation effect at large magnitudes ([Boore *et al.*, 2014](#)). A large scatter is observed in the residuals corrected for attenuation and site terms at low magnitudes ($M_w < 5.0$) and short periods ($T < 0.3$ s). At long periods (Fig. 5b), we do not observe negative slopes after M_h , and the variability is smaller.

Figure 6 shows the path term F_D , which includes the anelastic attenuation and the magnitude-dependent geometrical spreading, for two events of magnitudes 4 and 6. The p values relative to the distance coefficients are very low (Table 3), demonstrating the statistical importance of this term. The coefficient c_3 , relative to the anelastic attenuation, is also very well constrained. As expected, the ground motion attenuates faster at short periods.

The site term F_S is shown in Figure 7. The data points have been plotted in two different colors to evidence measured versus inferred V_{S30} . The site term is inversely proportional to V_{S30} , and as one could expect, records associated with measured V_{S30} show a lower scatter than the records with inferred V_{S30} . Most of the V_{S30} are inferred from the slope angle, and this can be a reason for an increase of the aleatory variability in the final model. However, restricting the dataset to the records associated with measured V_{S30} causes a significant improvement of sigma estimates, and on the other hand, a dramatic reduction the number of records (1/3), increasing the uncertainty about the estimation of the median ([Lanzano *et al.*, 2019](#)).

Analysis of Residuals

Figure 8 shows the between-event errors as a function of earthquake magnitude, the between-station errors with respect to V_{S30} , and the event- and site-corrected residuals as a function of the R_{JB} .

The between-event term is in the $[-0.4, 0.4]$ range in $\log_{10}$ units, with zero mean and no trend with magnitude. The standard deviation τ is equal to 0.18 at $T = 0.1$ s and 0.16 at $T = 10$ s. Two Japanese events (M_w 6.6 Fukuoka

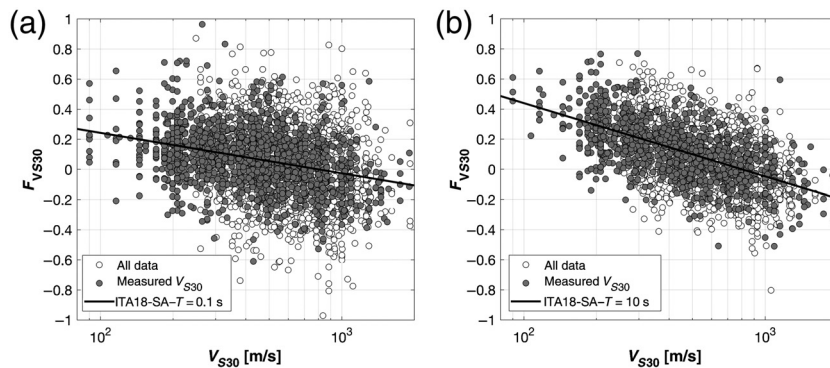


Figure 7. Site term as a function of V_{S30} against the residuals corrected for source and distance terms (gray circles are measured V_{S30} , and white circles are V_{S30} inferred from slope). (a) $T = 0.1$ s and (b) $T = 10$ s.

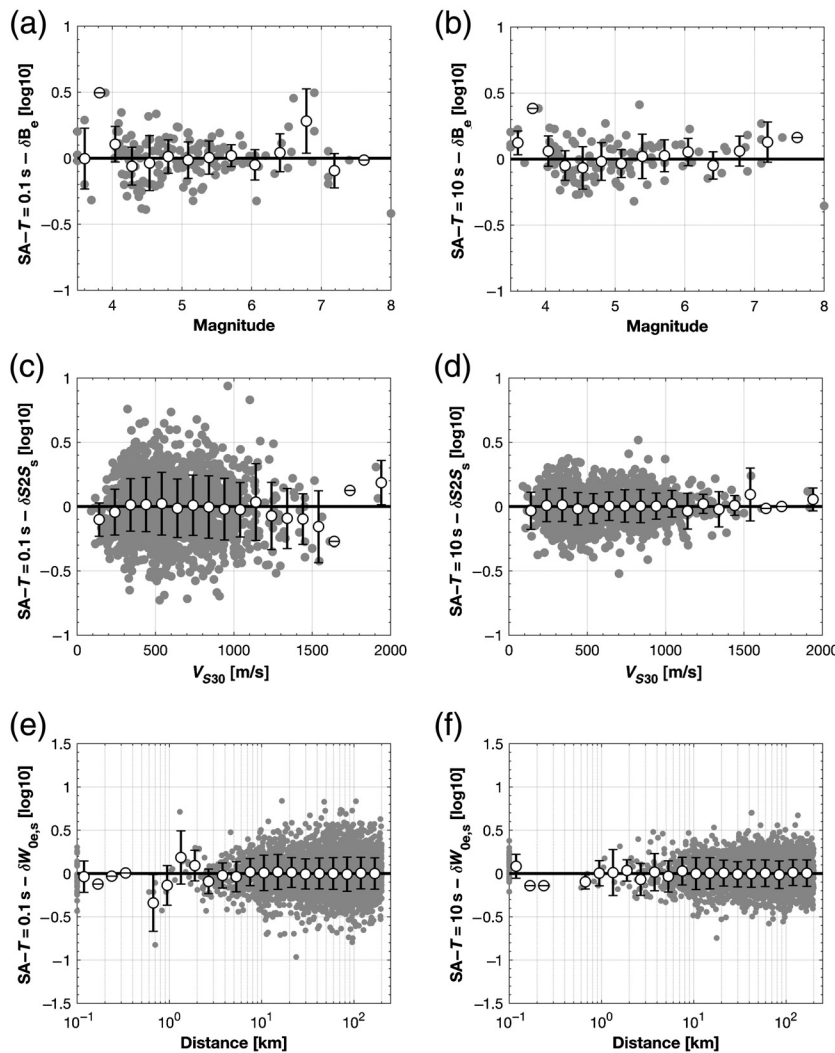


Figure 8. Residuals of the ITA18 model. (a,b) Between-event residuals versus magnitude, (c,d) site-to-site residuals versus V_{S30} , (e,f) event- and site-corrected residuals versus distance. (a,c,e) $T = 0.1$ s and (b,d,f) $T = 10$ s.

and M_w 6.9 Iwate earthquakes) are affected by large between-event errors at $T = 0.1$ s, and this could be ascribed to a different stress drop with respect to Italian events (Oth, 2013). A large negative between-event error is also related to the M_w 8.0 New Zealand event. We consider large between-event errors acceptable in case of strong global events because they occur in a different tectonic environment. On the other hand, they are fundamental to extend the range of validity of the model.

The site-to-site error is unbiased with respect to V_{S30} , especially in the range containing the largest number of records (e.g., 400–800 m/s). In general, the site-to-site error has the largest variability at $T = 0.1$ s ($\phi_{S2S} = 0.26$), probably because of the different high-frequency attenuation (κ) among sites with the same V_{S30} (Laurendeau *et al.*, 2013). The event- and site-corrected residuals show unbiased trends with distance; the standard deviations ϕ_0 are 0.21 at 0.1 s and 0.18 at 10 s, respectively.

Following Scasserra *et al.* (2009), the event- and site-corrected residuals $\delta W_{0e,s}$ are plotted in Figure 9 against the predicted PGA on rock sites ($V_{S30} = 800$ m/s) to explore possible nonlinear effects. The analysis is carried out for sites with $V_{S30} \leq 360$ m/s and two periods ($T = 0.1$ and 0.3 s). $\delta W_{0e,s}$ represents the error of each observed IMs with respect to the event- and site-corrected prediction. If a site is affected by nonlinear behavior, the residual will have negative values because a reduction of high-frequency ground motion is expected in case of strong shaking. A weak correlation is found for PGA larger than 200 cm/s² because the last bin is slightly negative. The records, supposedly affected by nonlinear soil effects, are only 86 (1.5% of the dataset) relative to 19 permanent and temporary strong-motion stations installed in Italy, most of which are located in the large alluvial basin of Po River (Northern Italy).

These evidences support the choice of not introducing the nonlinear site term in the functional form that, in this stage, would be constrained by very few data and would have a negligible impact both on median and standard deviation of the model.

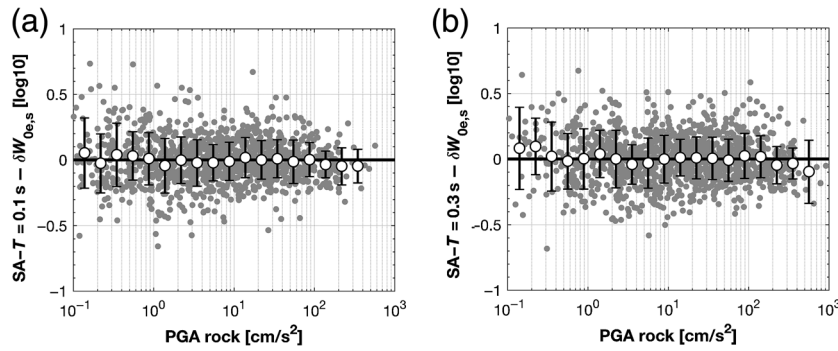


Figure 9. Event- and site-corrected residuals of sites with $V_{S30} \leq 360$ m/s against predicted peak ground acceleration (PGA) on rock ($V_{S30} = 800$ m/s). (a) $T = 0.1$ s and (b) $T = 0.3$ s.

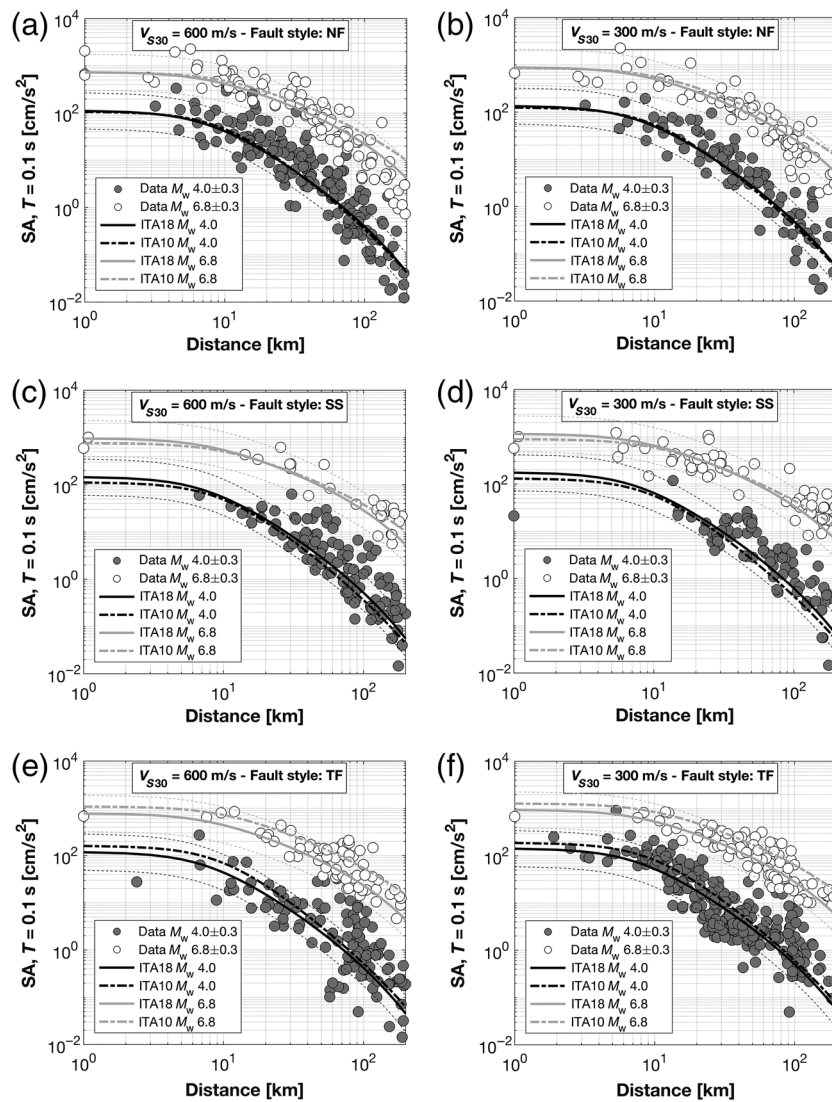


Figure 10. Predictions of ITA18 and ITA10 against observations at $T = 0.1$ s for M_w 4.0 and 6.8. (a) $V_{S30} = 600$ m/s and NF, (b) $V_{S30} = 300$ m/s and NF, (c) $V_{S30} = 600$ m/s and SS, (d) $V_{S30} = 300$ m/s and SS, (e) $V_{S30} = 600$ m/s and TF, and (f) $V_{S30} = 300$ m/s and TF. NF, normal fault; TF, thrust/reverse fault; SS, strike-slip fault.

Comparison with the Model by Bindi *et al.* (2011)

Figure 10 shows the observations against predictions of ITA18 and ITA10 for two magnitudes (M_w 4.0 and 6.8, which are close to the upper and lower limits of the dataset) and different SoF at $T = 0.1$ s. The comparison is carried out for two values of V_{S30} , 600 and 300 m/s that are representative of stiff and soft soils, respectively. The comparison is repeated in Figure 11 for $T = 1$ s. From a visual inspection, the observations are well represented by the ITA18 predictions.

At short periods, ITA10 and ITA18 predict similar amplitudes, except for TF events, for which ITA18 predicts lower values than ITA10. At intermediate periods, the largest differences are found for M_w 6.8, in which ITA10 predictions are higher than ITA18. This is expected because ITA10 is poorly constrained at $M_w > 6.5$, and this can be the cause of the overprediction. The anelastic attenuation term is better represented in ITA18 than ITA10, although at large magnitudes, an overprediction for NF events (white circles) still exists at distances higher than 80 km. The poor performance of the anelastic attenuation of ITA10 was also highlighted by several authors (Stewart *et al.*, 2012; Luzi, Pacor, Puglia, *et al.*, 2017; Zimmario *et al.*, 2018).

The percentage difference in the predictions between ITA18 and ITA10 is shown in Table 4 over all IMs (PGA, PGV, and SA in the 0.04- to 2-s period range) and for different values of magnitude, distance, V_{S30} , and SoF.

Because ITA10 has been calibrated for soil categories, we can only compare the ITA10 EC8 coefficients with the ITA18 site terms evaluated at single V_{S30} values representative of the soil classes. In particular, EC8 classifies the sites in five categories based on V_{S30} intervals: EC8-A $V_{S30} > 800$ m/s; EC8-B $V_{S30} = 360$ –800 m/s; EC8-C $V_{S30} = 180$ –360 m/s; EC8-D $V_{S30} < 180$ m/s; and EC8-E corresponds to 5–20 m EC8-C or EC8-D sites underlain by rock (with shear-wave velocity > 800 m/s). Figure 12 shows that the trend of the EC8-C coefficient fits the ITA18 site term evaluated at $V_{S30} = 250$ m/s.

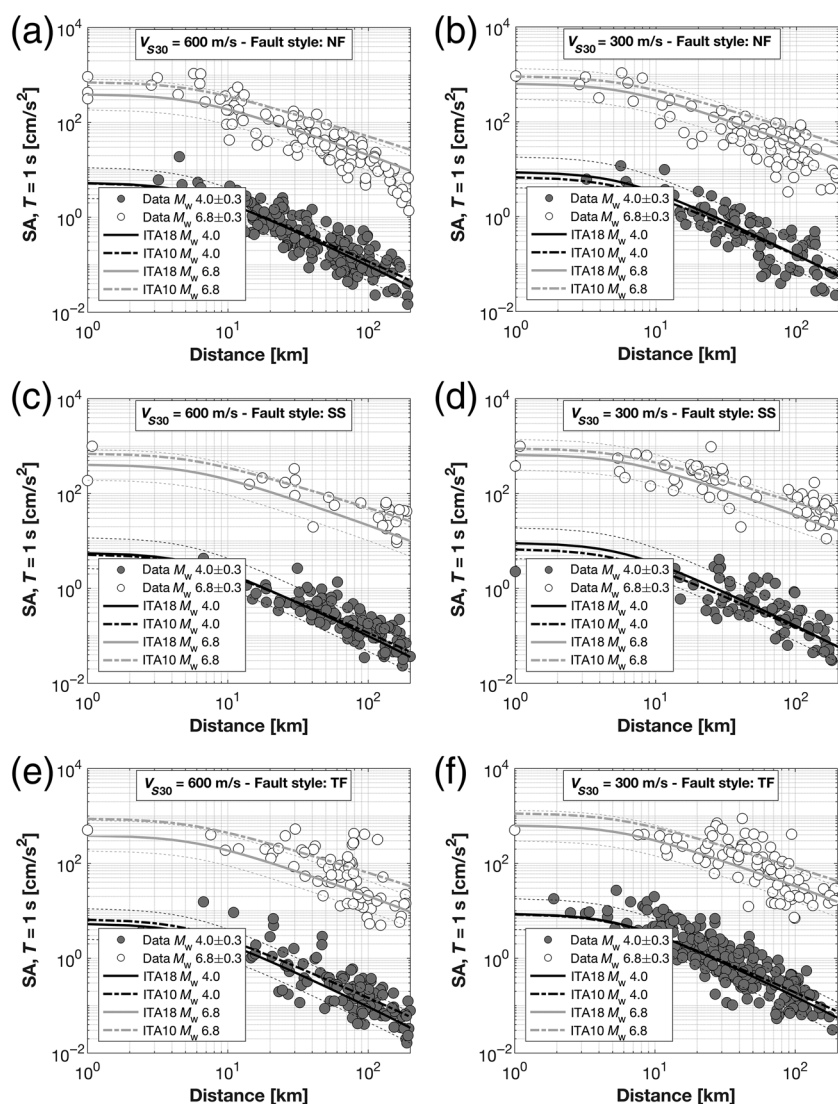


Figure 11. Predictions of ITA18 and ITA10 against observations at $T = 1.0$ s for M_w 4.0 and 6.8. (a) $V_{S30} = 600$ m/s and NF, (b) $V_{S30} = 300$ m/s and NF, (c) $V_{S30} = 600$ m/s and SS, (d) $V_{S30} = 300$ m/s and SS, (e) $V_{S30} = 600$ m/s and TF, and (f) $V_{S30} = 300$ m/s and TF. NF, normal fault; TF, thrust/reverse fault; SS, strike-slip fault.

Whereas the EC8-B coefficient is between the site terms estimated at $V_{S30} = 250$ m/s and $V_{S30} = 600$ m/s, the EC8-D trend is very peak shaped around $T = 1$ s and does not fit the ITA18 site term estimated at $V_{S30} = 100$ m/s. This could be attributed to the limited number of records of EC8-D category used to derive ITA10.

Figure 13 compares the standard deviations (between-event, between-station, and total standard deviation) of ITA10 and ITA18. The general trend is a decrease of the total sigma of ITA18, for periods larger than 0.1 s. In particular, the component showing the larger decrease is the between-event, attributable to the improved accuracy in the location and magnitude of events, because of the enhancement of monitoring networks and quality of instruments. We do not observe differences in the site-to-site standard deviation of the two models at short periods because the adopted site parameterization is still not efficient in reducing the variability.

Modeling the Uncertainties

Aleatory Uncertainty

We explore the possibility to build up a heteroscedastic model for the different components of the variability of ITA18 model. The magnitude-dependent model, selected for the event- and site-corrected variability (ϕ_0), is:

Table 4
Percentage Differences between ITA18 and ITA10 Computed over All Intensity Measures (IMs)

Joyner-Boore Distance (R_{JB}) (km)	Focal Mechanism	Percentage Increment ($\Delta\%$)					
		$V_{S30} = 800$ m/s			$V_{S30} = 350$ m/s		
		M_w 4.0	M_w 5.5	M_w 7.0	M_w 4.0	M_w 5.5	M_w 7.0
1	Normal faulting	40.1	54.4	15.2	15.4	27.6	-5.2
50		30.4	24.9	-19.0	7.1	3.1	-33.4
200		21.7	2.0	-43.0	0.2	-15.4	-52.8
1	Strike-slip faulting	60.0	75.8	32.1	31.3	44.8	8.2
50		49.2	42.2	-7.2	22.1	17.0	-24.0
200		39.0	15.4	-35.3	14.0	-4.6	-46.6
1	Reverse faulting	2.5	12.9	-16.0	-15.5	-6.6	-30.9
50		-4.7	-8.7	-41.0	-21.6	-24.6	-51.4
200		-11.3	-25.6	-58.4	-26.9	-38.2	-65.5

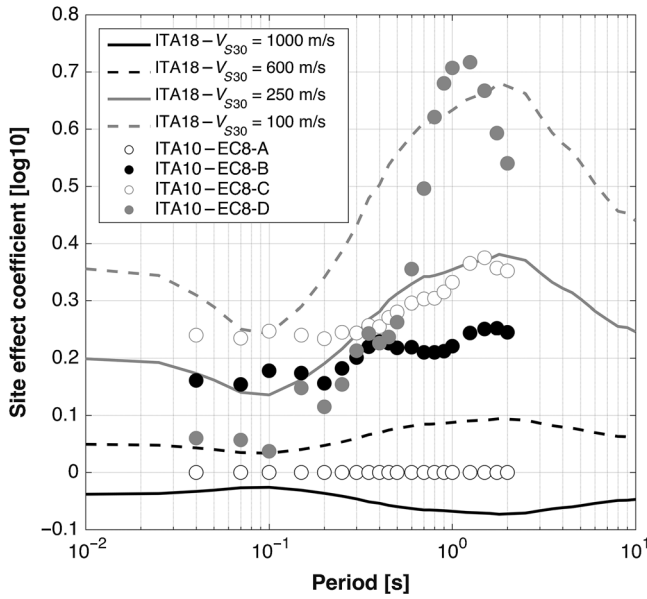


Figure 12. Comparison between ITA18 site terms and ITA10 coefficients relative to Eurocode 8 (EC8) categories (A–D).

$$\phi_0(M_w) = \begin{cases} \phi_{0,1} & \text{where } M_w \leq M_1 \\ \phi_{0,1} + (\phi_{0,2} - \phi_{0,1})(M_w - M_1) & \text{where } M_1 < M_w < M_2, \\ \phi_{0,2} & \text{where } M_w \geq M_2 \end{cases} \quad (7)$$

in which $\phi_{0,1}$ and $\phi_{0,2}$ represent the aleatory variability at small and large magnitudes, respectively, and M_1 and M_2 are the corner magnitudes. After some trial tests, we assume M_1 dependent on IMs, ranging from 4.5 to 5.5, and M_2 constant and equal to 6.0. At periods longer than 2 s, the dependence of ϕ_0 on magnitude is less clear, and the differences with respect to the homoscedastic models can be considered negligible. We provide examples of the ϕ_0 model (equation 7) for $T = 0.1$ and 1 s in Figure 14a and 14b, respectively. The standard deviations are binned and plotted with their 95% confidence interval: the error of sigma in each magnitude bin is small, except for the M_w 7.0 bin, because of the limited amount of data at large magnitudes.

We also develop a model for ϕ_{S2S} as a function of V_{S30}

$$\phi_{S2S}(V_{S30}) = \begin{cases} \phi_{S2S,1} & \text{where } V_{S30} \leq V_1 \\ \phi_{S2S,1} + (\phi_{S2S,2} - \phi_{S2S,1}) \left(\frac{\log_{10}(V_2/V_{S30})}{\log_{10}(V_2/V_1)} \right) & \text{where } V_1 < V_{S30} < V_2, \\ \phi_{S2S,2} & \text{where } V_{S30} \geq V_2 \end{cases} \quad (8)$$

in which the corner velocities are $V_1 = 250$ m/s and $V_2 = 400$ m/s at all IMs. Figure 14c,d gives an example of the ϕ_{S2S} model: the errors of ϕ_{S2S} bins are larger than ϕ_0 , especially at the highest and lowest V_{S30} bins.

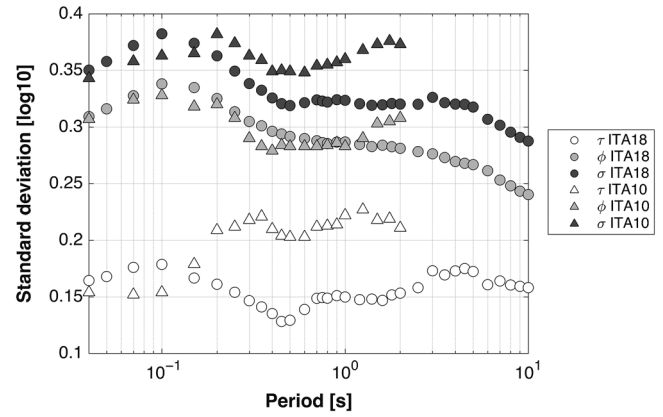


Figure 13. Standard deviations of ITA18 and ITA10. σ (total sigma, dark gray), τ (between-event sigma, white), and ϕ (within-event sigma, light gray).

Finally, we decide not to build a heteroscedastic model for τ , also because this component is already smaller than those observed in other predictive models (e.g., Boore *et al.*, 2014).

In Table S2, we provided the values of $\phi_{0,1}$, $\phi_{0,2}$, M_1 , M_2 , $\phi_{S2S,1}$, and $\phi_{S2S,2}$.

Epistemic Uncertainty

The epistemic uncertainty can be partially quantified by estimating the statistical uncertainty in the median predictions, calculated on the model fit and the data distribution. For a set of predictor variables (location x_0), the epistemic uncertainty can be calculated as (Al Atik and Youngs, 2014; Bindi *et al.*, 2017):

$$\text{var}[\log_{10} Y]_{x_0} = J_0^T [\text{varCov}_{x_i}] J_0, \quad (9)$$

in which x_i are the data points used to develop the model; J_0 is the Jacobian matrix, that is, the gradient of the model with respect to its coefficients, evaluated in the predictive location x_0 ; and Cov_{x_i} is the variance–covariance matrix of the coefficients, evaluated at all data points x_i . The variance–covariance matrix quantifies the uncertainty in the coefficients (diagonal elements) and the trade-offs between pair of coefficients (off-diagonal elements).

The square root of the left side term in equation (9) is the standard deviation of the median σ_μ that quantifies the epistemic uncertainty in the median estimation due to the combined effects of limited data and unknown effects that are not accounted for in the functional form.

Figure 15 shows σ_μ with respect to the four predictive variables (magnitude, distance, V_{S30} , and SoF) at different spectral periods. The largest variation on the median uncertainty depends on the earthquake magnitude. Low σ_μ values are observed in the magnitude range where the data sampling is the largest (M_w in the 4.5–6.5 range), but σ_μ increases at magnitude greater than 7.0, where the data sampling is poorer. $T = 0.1$ s is the period at which the higher

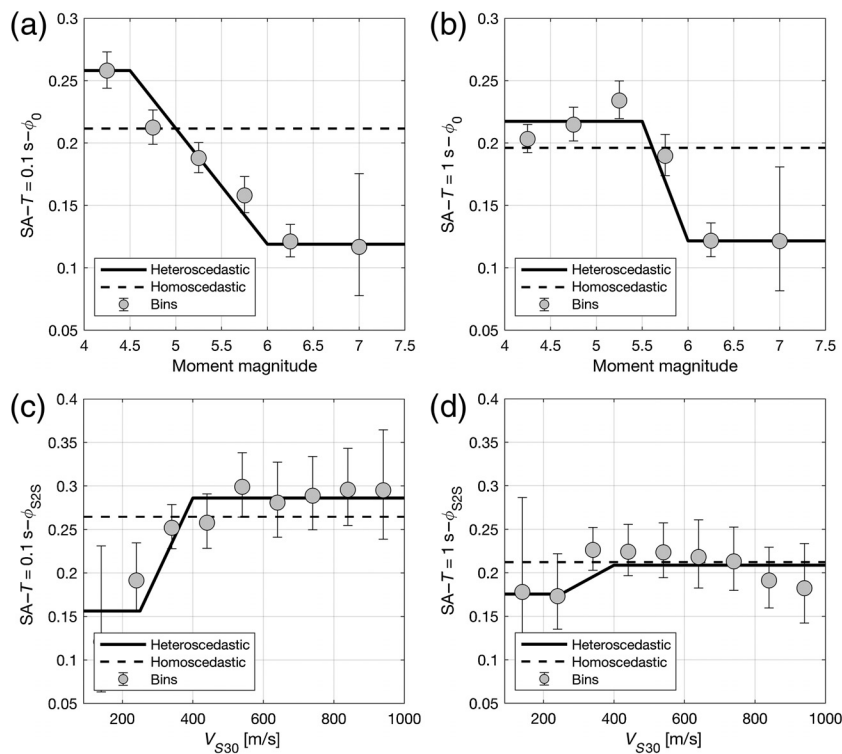


Figure 14. Heteroscedastic models for aleatory variability. (a) Event- and site-corrected standard deviation (ϕ_0) as a function of magnitude at (a) $T = 0.1$ s and (b) $T = 1.0$ s; site-to-site variability (ϕ_{S2S}) as a function of V_{S30} at (c) $T = 0.1$ s and (d) $T = 1.0$ s.

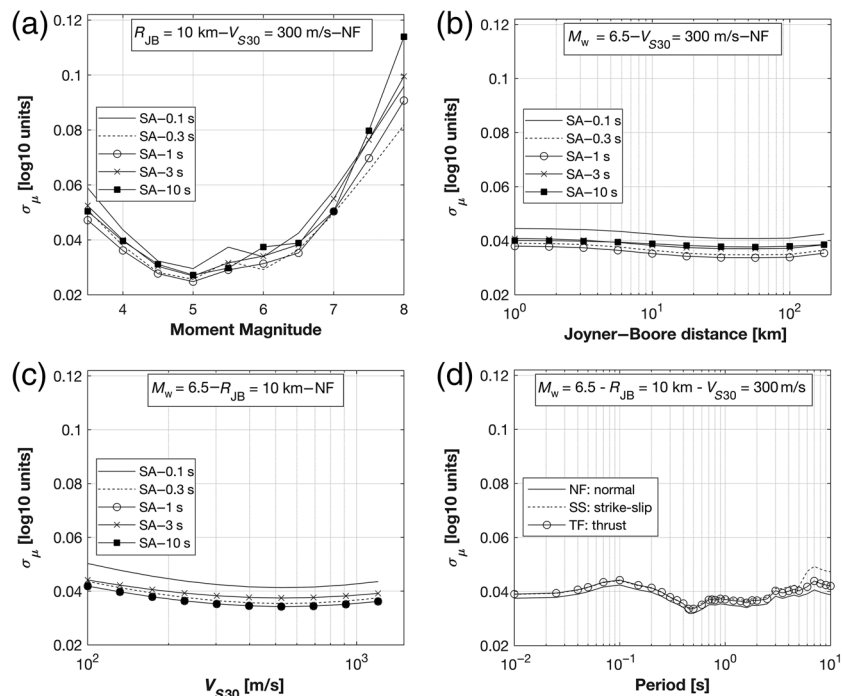


Figure 15. Epistemic uncertainty σ_μ as a function of predictive variables. (a) Moment magnitude M_w , (b) Joyner-Boore distance R_{JB} , (c) V_{S30} , (d) SoF.

uncertainty about the median is observed, and, according to this study, is the period characterized by the largest variability of the observations. The values of σ_μ are provided in Table S3 for each IMs and different magnitudes.

Conclusions

The research described in this article consists in a revision of the ITA10 ground-motion prediction model developed by Bindi *et al.* (2011) for shallow crustal earthquakes in Italy. Although its overall good performance, even for the general active shallow crust in Europe, ITA10 has several restrictions because of the limited number of data available at the time it was calibrated (e.g., 2009). In particular, ITA10 has M_w 6.9 as maximum usable magnitude and 2 s as maximum usable period for the GMPEs.

In this work, we recalibrated the set of GMPEs for Italy, named ITA18, performing a regression on a much wider dataset (5607 records, 146 events, and 1657 stations), including regional and very well-sampled worldwide events, to extend the magnitude range. Waveforms characterized by moment magnitude, distance, and site information are only included in the calibration dataset. We prefer to keep as much data as possible, including the records from stations with V_{S30} inferred by proxies to have a robust estimation of source scaling and attenuation with distance. This choice causes an increase of aleatory variability while reducing the epistemic uncertainty.

We approach the regression in two steps: in the first step, we constrain three variables—that is, the hinge and the reference magnitude and the earthquake pseudodepth—through a nonlinear regression. In the second step, we performed mixed-effect linear regression to obtain the coefficients and the standard deviations of the between-event, site-to-site, and event-and-site corrected residuals. A heteroscedastic model is proposed for the event- and site-corrected and site-to-site variabilities, similar to those obtained in NGA-West2 GMPEs. In addition, the epistemic uncertainty associated with the median prediction is provided as a contribution for the

construction of logic trees in probabilistic seismic hazard studies.

For the reasons we expose, the proposed model can supersede the Bindi *et al.* (2011) for the prediction of the horizontal ground motion. ITA18 is calibrated both for rupture and R_{JB} and has the following characteristics:

- moment magnitude range: 4.0–8.0;
- distance range: 0–200 km;
- event depth: shallower than 30 km;
- IMs (RotD50): SA at 5% damping for 36 periods in the 0.04–10 s range, PGA and PGV;
- site term: linear function of V_{S30} ;
- aleatory variability: partition of the variability in between-event, site-to-site, and event- and site-corrected variability; heteroscedastic model for the site-to-site and event- and site-corrected variabilities;
- epistemic variability: uncertainty in the median prediction as a function of IM and moment magnitude.

We decide to keep the site effects terms in the functional form as simple as possible in this GMPEs release, including only the linear dependence on V_{S30} . Additional studies are necessary to introduce an effective classification scheme, able to reduce the aleatory variability, incorporating both the linear and nonlinear effects. Because the V_{S30} can be considered as a rough proxy even for the linear site effects (Castellaro *et al.*, 2008), additional or different explanatory variables such as the predominant period of the site, the depth of the bedrock, or the kappa parameter should be accounted for. Moreover, because the observational data are very few, the nonlinear effects could be supported by numerical simulation of the seismic site response, similar to that was done in the framework of the NGA-West2 project.

Data and Resources

All data used in this article came from published sources listed in the references. The accelerometric waveforms used in this study are published in the following databases: Engineering Strong-Motion (ESM, <https://www.orfeus-eu.org/data/strong>, last accessed January 2018), Italian ACcelerometric Archive (ITACA, <http://itaca.mi.ingv.it>, last accessed August 2018), Kiknet (<http://www.kyoshin.bosai.go.jp>, last accessed August 2018), Consortium of Organizations for Strong Motion Observation Systems/Center for Engineering Strong-Motion Data (COSMOS/CESMD, <https://www.strongmotioncenter.org>, last accessed January 2019), Internet-Site for European Strong-Motion Data (<http://www.isesd.hi.is>, last accessed August 2018) only for Icelandic data, Building & Housing Research Center of Iran (<http://smd.bhrc.ac.ir/portal/en>, last accessed August 2018), Disaster and Emergency Management Presidency of Turkey (<http://kyhdata.deprem.gov.tr>, last accessed August 2018), and GeoNet (<https://www.geonet.org.nz/data/tools/FDSN>, last accessed August 2018).

Acknowledgments

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References

- Abrahamson, N. A., W. J. Silva, and R. Kamai (2014). Summary of the ASK14 ground motion relation for active crustal regions, *Earthq. Spectra* **30**, no. 3, 1025–1055.
- Al Atik, L., and R. R. Youngs (2014). Epistemic uncertainty for NGA-West2 models, *Earthq. Spectra* **30**, no. 3, 1301–1318.
- Al Atik, L., N. A. Abrahamson, J. J. Bommer, F. Scherbaum, F. Cotton, and N. Kuehn (2010). The variability of ground-motion prediction models and its components, *Seismol. Res. Lett.* **81**, no. 5, 794–801.
- Ambraseys, N. N., J. Douglas, S. K. Sarma, and P. M. Smit (2005). Equations for the estimation of strong ground motions from shallow crustal earthquakes using data from Europe and the Middle East: Horizontal peak ground acceleration and spectral acceleration, *Bull. Earthq. Eng.* **3**, no. 1, 1–53.
- Anderson, J. G., and J. N. Brune (1999). Probability seismic hazard analysis without the ergodic assumption, *Seismol. Res. Lett.* **70**, 19–28.
- Atkinson, G. M. (2006). Single-station sigma, *Bull. Seismol. Soc. Am.* **96**, 446–455.
- Atkinson, G. M., and J. Adams (2013). Ground motion prediction equations for application to the 2015 Canadian National Seismic Hazard Maps, *Can. J. Civ. Eng.* **40**, no. 10, 988–998.
- Bates, D., M. Mächler, B. Bolker, and S. Walker (2015). Fitting linear mixed-effects models using lme4, *J. Stat. Software* **67**, no. 1, 1–48.
- Bindi, D., F. Cotton, S. R. Kotha, C. Bosse, D. Stromeyer, and G. Grünthal (2017). Application-driven ground motion prediction equation for seismic hazard assessments in non-cratonic moderate-seismicity areas, *J. Seismol.* **21**, no. 5, 1201–1218.
- Bindi, D., F. Pacor, L. Luzi, R. Puglia, M. Massa, G. Ameri, and R. Paolucci (2011). Ground motion prediction equations derived from the Italian strong-motion database, *Bull. Earthq. Eng.* **9**, 1899–1920.
- Bommer, J. J., J. Douglas, and F. O. Strasser (2003). Style-of-faulting in ground-motion prediction equations, *Bull. Earthq. Eng.* **1**, 171–203.
- Bommer, J. J., P. J. Stafford, J. E. Alarcón, and S. Akkar (2007). The influence of magnitude range on empirical ground-motion prediction, *Bull. Seismol. Soc. Am.* **97**, no. 6, 152–170.
- Boore, D. M. (2010). Orientation-independent, nongeometric-mean measures of seismic intensity from two horizontal components of motion, *Bull. Seismol. Soc. Am.* **100**, no. 4, 1830–1835.
- Boore, D. M., and G. M. Atkinson (2008). Ground-motion prediction equations for the average horizontal component of PGA, PGV, and 5%-damped PSA at spectral periods between 0.01 s and 10.0 s, *Earthq. Spectra* **24**, no. 1, 99–138.
- Boore, D. M., and J. J. Bommer (2005). Processing of strong-motion accelerograms: Needs, options and consequences, *Soil Dynam. Earthq. Eng.* **25**, 93–115.
- Boore, D. M., J. P. Stewart, E. Seyhan, and G. M. Atkinson (2014). NGA-West2 equations for predicting PGA, PGV, and 5% damped PSA for shallow crustal earthquakes, *Earthq. Spectra* **30**, no. 3, 1057–1085.

- Bozorgnia, Y., N. A. Abrahamson, L. Al-Atik, T. D. Ancheta, G. M. Atkinson, J. W. Baker, A. Baltay, D. M. Boore, K. W. Campbell, B. S.-J. Chiou, *et al.* (2014). NGA-West2 research project, *Earthq. Spectra* **30**, no. 3, 973–987.
- Campbell, K. W., and Y. Bozorgnia (2014). NGA-West2 ground motion model for the average horizontal components of PGA, PGV, and 5% damped linear acceleration response spectra, *Earthq. Spectra* **30**, no. 3, 1087–1115.
- Castellaro, S., F. Mulargia, and P. L. Rossi (2008). V_{S30} : Proxy for seismic amplification?, *Seismol. Res. Lett.* **79**, no. 4, 540–543.
- Chiou, B. S.-J., and R. R. Youngs (2014). Update of the Chiou and Youngs NGA model for the average horizontal component of peak ground motion and response spectra, *Earthq. Spectra* **30**, no. 3, 1117–1153.
- D'Amico, M., C. Felicetta, E. Schiappapietra, F. Pacor, F. Gallovič, R. Paolucci, R. Puglia, G. Lanzano, S. Sgobba, and L. Luzi (2018). Fling effects from near-source strong-motion records: Insights from M_w 6.5, 2016, Norcia earthquake (central Italy), *Seismol. Res. Lett.* doi: [10.1785/0220180169](https://doi.org/10.1785/0220180169).
- Delavaud, E., F. Cotton, S. Akkar, F. Scherbaum, L. Danciu, C. Beauval, S. Drouet, J. Douglas, R. Basili, M. A. Sandikkaya, *et al.* (2012). Toward a ground-motion logic tree for probabilistic seismic hazard assessment in Europe, *J. Seismol.* **16**, no. 3, 451–473.
- Ditommaso, R., M. Mucciarelli, M. R. Gallipoli, and F. C. Ponso (2010). Effect of a single vibrating building on free-field ground motion: Numerical and experimental evidences, *Bull. Earthq. Eng.* **8**, no. 3, 693–703.
- Douglas, J. (2018). Capturing geographically-varying uncertainty in earthquake ground motion models or what we think we know may change, in *Recent Advances in Earthquake Engineering in Europe. ECEE 2018*, K. Pitilakis (Editor), Geotechnical, Geological and Earthquake Engineering, Vol. 46, Springer, Cham, Switzerland.
- Ekström, G., M. Nettles, and A. M. Dziewonski (2012). The global CMT project 2004–2010: Centroid-moment tensors for 13,017 earthquakes, *Phys. Earth Planet. In.* **1**, no. 9, 200–201.
- Eurocode 8 (2003). Design of structures for earthquake resistance—Part 1: General rules seismic actions and rules for buildings, *EN 1998-1*, European Committee for Standardization, Brussels, Belgium.
- Felicetta, C., M. D'Amico, G. Lanzano, R. Puglia, E. Russo, and L. Luzi (2017). Site characterization of Italian accelerometric stations, *Bull. Earthq. Eng.* **15**, no. 6, 2329–2348.
- Grünthal, G., and R. Wahlström (2012). The European-Mediterranean Earthquake Catalogue (EMEC) for the last millennium, *J. Seismol.* **16**, no. 3, 535–570.
- Joyner, W. B., and D. M. Boore (1981). Peak horizontal acceleration and velocity from strong-motion records including records from the 1979 Imperial Valley, California earthquake, *Bull. Seismol. Soc. Am.* **71**, 2011–2038.
- Kamai, R., N. Abrahamson, and R. Graves (2014). Adding fling effects to processed ground-motion time histories, *Bull. Seismol. Soc. Am.* **104**, no. 4, 1914–1929.
- Kamai, R., N. Abrahamson, and W. J. Silva (2014). Nonlinear horizontal site amplification for constraining the NGA-West2 GMPEs, *Earthq. Spectra* **30**, no. 3, 1223–1240.
- Kulkarni, R. B., R. R. Youngs, and K. J. Coppersmith (1984). Assessment of confidence intervals for results of seismic hazard analysis, *Proc. of 8th World Conf. on Earthquake Engineering*, Vol. 1, 263–270.
- Lanzano, G., M. D'Amico, C. Felicetta, R. Puglia, L. Luzi, F. Pacor, and D. Bindi (2016). Ground-motion prediction equations for region-specific probabilistic seismic-hazard analysis, *Bull. Seismol. Soc. Am.* **106**, no. 1, 73–92.
- Lanzano, G., L. Luzi, F. Pacor, R. Puglia, C. Felicetta, M. D'Amico, and S. Sgobba (2019). Update of the ground motion prediction equations for Italy, *Proc. of the 7th International Conf. on Earthquake Geotechnical Engineering*, Rome, Italy, 17–20 June 2019.
- Lanzano, G., S. Sgobba, L. Luzi, R. Puglia, F. Pacor, C. Felicetta, M. D'Amico, F. Cotton, and D. Bindi (2018). The pan-European Engineering Strong-motion (ESM) flatfile: Compilation criteria and data statistics, *Bull. Earthq. Eng.* 1–22, doi: [10.1007/s10518-018-0480-z](https://doi.org/10.1007/s10518-018-0480-z).
- Laurendeau, A., F. Cotton, O. J. Ktenidou, L. F. Bonilla, and F. Hollender (2013). Rock and stiff-soil site amplification: Dependency on V_{S30} and κ_0 , *Bull. Seismol. Soc. Am.* **103**, no. 6, 3131–3148.
- Luzi, L., F. Pacor, and R. Puglia (2017). *Italian Accelerometric Archive*, v. 2.3, Dipartimento della Protezione Civile Nazionale, Istituto Nazionale di Geofisica e Vulcanologia, doi: [10.13127/ITACA.2.3](https://doi.org/10.13127/ITACA.2.3).
- Luzi, L., F. Pacor, R. Puglia, G. Lanzano, C. Felicetta, M. D'Amico, A. Michelini, L. Faenza, V. Lauciani, I. Iervolino, *et al.* (2017). The central Italy seismic sequence between August and December 2016: Analysis of strong-motion observations, *Seismol. Res. Lett.* **88**, 1219–1231.
- Luzi, L., R. Puglia, E. Russo, M. D'Amico, C. Felicetta, F. Pacor, G. Lanzano, U. Ceken, J. Clinton, G. Costa, *et al.* (2016). The engineering strong-motion database: A platform to access pan-European accelerometric data, *Seismol. Res. Lett.* **87**, no. 4, 987–997.
- Luzi, L., R. Puglia, E. Russo, and ORFEUS WG5 (2016). *Engineering Strong-Motion Database, version 1.0*, Observatories & Research Facilities for European Seismology, Istituto Nazionale di Geofisica e Vulcanologia, doi: [10.13127/ESM](https://doi.org/10.13127/ESM).
- Norme Tecniche per le Costruzioni (NTC) (2018). Aggiornamento delle Norme tecniche per le costruzioni, decreto 17-1-2018, *Gazzetta Ufficiale* **42**, 20-02-2018, Ordinary Suppl. n. 8 (in Italian).
- Oth, A. (2013). On the characteristics of earthquake stress release variations in Japan, *Earth Planet. Sci. Lett.* **377**, 132–141.
- Pacor, F., R. Paolucci, L. Luzi, F. Sabetta, A. Spinelli, A. Gorini, M. Nicoletti, S. Marcucci, L. Filippi, and M. Dolce (2011). Overview of the Italian strong motion database ITACA 1.0, *Bull. Earthq. Eng.* **9**, no. 6, 1723–1739.
- Paolucci, R., F. Pacor, R. Puglia, G. Ameri, C. Cauzzi, and M. Massa (2011). Record processing in ITACA, the new Italian strong-motion database, in *Earthquake Data in Engineering Seismology*, S. Akkar, P. Gulkan, and T. Van Eck (Editors), Geotechnical, Geological and Earthquake Engineering Series, Vol. 14, Springer, Dordrecht, The Netherlands, 99–113.
- Pondrelli, S., and S. Salimbeni (2015). Regional moment tensor review: An example from the European–Mediterranean region, in *Encyclopedia of Earthquake Engineering*, M. Beer, I. A. Kougiumtzooglou, E. Patelli, and I. S.-K. Au (Editors), Springer, Berlin, Germany, 1–15, available at http://link.springer.com/referenceworkentry/10.1007/978-3-642-36197-5_301-1.
- Rodriguez-Marek, A., G. A. Montalva, F. Cotton, and F. Bonilla (2011). Analysis of single-station standard deviation using the KiK-net data, *Bull. Seismol. Soc. Am.* **101**, 1242–1258.
- Sadigh, K., C.-Y. Chang, J. A. Egan, F. Makdisi, and R. R. Youngs (1997). Attenuation relationships for shallow crustal earthquakes based on California strong motion data, *Seismol. Res. Lett.* **68**, no. 1, 180–189.
- Scasserra, G., J. P. Stewart, P. Bazzurro, G. Lanzo, and F. Mollaioli (2009). A comparison of NGA ground-motion prediction equations to Italian data, *Bull. Seismol. Soc. Am.* **99**, no. 5, 2961–2978.
- Scognamiglio, L., E. Tinti, and A. Michelini (2009). Real-time determination of seismic moment tensor for the Italian region, *Bull. Seismol. Soc. Am.* **99**, no. 4, 2223–2242.
- Stewart, J. P., K. Afshari, and C. A. Goulet (2017). Non-ergodic site response in seismic hazard analysis, *Earthq. Spectra* **33**, 1385–1414.
- Stewart, J. P., G. Lanzo, A. Pagliaroli, G. Scasserra, G. Di Capua, S. Peppoloni, R. B. Darragh, and M. Gregor (2012). Ground motion recordings from the M_w 6.3 2009 L'Aquila earthquake in Italy and their engineering implications, *Earthq. Spectra* **28**, 317–345.

- Strasser, F. O., N. A. Abrahamson, and J. J. Bommer (2009). Sigma: Issues, insights, and challenges, *Seismol. Res. Lett.* **80**, no. 1, 40–56.
- Toro, G., N. Abrahamson, and J. Schneider (1997). Letter to the editor, *Seismol. Res. Lett.* **68**, no. 3, 481–482.
- Wald, D. J., and T. I. Allen (2007). Topographic slope as a proxy for seismic site conditions and amplification, *Bull. Seismol. Soc. Am.* **97**, no. 5, 1379–1395.
- Wasserstein, R. L., and N. A. Lazar (2016). The ASA's statement on p-values: Context, process, and purpose, *Am. Stat.* **70**, no. 2, 129–133.
- Zimmaro, P., G. Scasserra, J. P. Stewart, T. Kishida, G. Tropeano, M. Castiglia, and P. Pelekis (2018). Strong ground motion characteristics from 2016 Central Italy earthquake sequence, *Earthq. Spectra* **34**, no. 4, 1611–1637, doi: [10.1193/091817EQS184M](https://doi.org/10.1193/091817EQS184M).

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